

1. Unit weight of fill of 21.2 kN/m³, stope height 50 m, thickness of stope as 20 m, stope width along strike 25 m, fill friction angle as 32 degree, using Mitchell et al. (1982) solution determine the required strength of backfill, consider FS1.

- **Unit weight of fill (γ):** 21.2 kN/m³
- **Stope height (H):** 50 m
- **Thickness of stope (L , perpendicular to strike):** 20 m
- **Fill friction angle (ϕ):** 32°
- **Factor of Safety (FS):** 1

(Note: The stope width along the strike, $B = 25$ m, is provided but cancels out in the final cohesion and UCS equations for an FS of 1.)

1. Required Cohesion (c)

The Mitchell et al. (1982) formula for the required apparent cohesion of the backfill on the failure plane is :

$$c = \frac{\gamma H}{2\left(\frac{H}{L} + \tan \alpha\right)}$$

Where α (the critical sliding plane angle) is calculated as :

$$\alpha = 45^\circ + \frac{\phi}{2} = 45^\circ + \frac{32^\circ}{2} = 61^\circ$$

Plugging in the given values:

$$c = \frac{21.2 \times 50}{2\left(\frac{50}{20} + \tan 61^\circ\right)}$$

$$c = \frac{1060}{2(2.5 + 1.804)}$$

$$c = \frac{1060}{8.608} \approx 123.14 \text{ kPa}$$

2. Required Unconfined Compressive Strength (UCS)

The Mitchell et al. (1982) formula used to determine the required UCS assumes a simplified, frictionless backfill condition ($\phi = 0$) ³ ⁴. The equation is:

$$UCS = \frac{\gamma H}{1 + \frac{H}{L}}$$

Plugging in the given values:

$$UCS = \frac{21.2 \times 50}{1 + \frac{50}{20}}$$

$$UCS = \frac{1060}{1 + 2.5}$$

$$UCS = \frac{1060}{3.5} \approx \checkmark 302.86 \text{ kPa}$$

2. For the problem numerical in question 1 for FS 2 calculate the required strength.

Given Parameters:

- Unit weight of fill (γ): 21.2 kN/m³
- Stope height (H): 50 m
- Thickness of stope (L , perpendicular to strike): 20 m
- Fill friction angle (ϕ): 32°
- Factor of Safety (FS): 2

(Note: As with the FS = 1 calculation, the stope width along the strike, $B = 25$ m, is not needed because it cancels out in the final cohesion equation ² ³.)

1. Calculate the Critical Sliding Plane Angle (α)

The critical sliding plane angle is calculated using the internal friction angle (ϕ) of the backfill ³ ⁴ :

$$\alpha = 45^\circ + \frac{\phi}{2}$$

2. Calculate the Modified Factor (A)

To modify the Mitchell equation for an FS of 2, we first need to calculate an intermediate parameter A ² :

$$A = \frac{1}{\sin(2\alpha) - \cos^2(\alpha) \tan(\phi)}$$

Plugging in our values ($\alpha = 61^\circ$, $\phi = 32^\circ$):

- $\sin(2 \times 61^\circ) = \sin(122^\circ) \approx 0.848$
- $\cos^2(61^\circ) \approx (0.4848)^2 \approx 0.235$
- $\tan(32^\circ) \approx 0.6249$

$$A = \frac{1}{0.848 - (0.235 \times 0.6249)} = 1.426$$

3. Required Cohesion (c) for FS = 2

Using the modified Mitchell equation for an FS of 2 :

$$c = \frac{\gamma H}{2\left(\frac{H}{L}\right) + A}$$

$$c = \frac{21.2 \times 50}{2\left(\frac{50}{20}\right) + 1.426}$$

$$c = \frac{1060}{2(2.5) + 1.426}$$

$$c = \frac{1060}{5 + 1.426} = 164.95 \text{ kPa}$$

4. Required Unconfined Compressive Strength (UCS)

To find the UCS based on the Coulomb criterion (using both cohesion and the friction angle), we use the following formula ⁵ ⁶ :

$$UCS = 2c \tan\left(45^\circ + \frac{\phi}{2}\right)$$

$$UCS = 2(164.95) \tan(61^\circ)$$

$$UCS = 329.9 \times 1.804$$

$$UCS \approx 595.15 \text{ kPa}$$

3. Unit weight of fill of 21.2 kN/m³, stope height 50 m, thickness of stope as 20 m, stope width along strike 25, fill friction angle as 32 degree, adherence ratio as 0.5, no overlying load, using Li & Aubertin solution determine the required strength of backfill, consider FS 1.

- Unit weight of fill (γ): 21.2 kN/m³
- Stope height (H): 50 m
- Thickness of stope (L , perpendicular to strike): 20 m
- Stope width along strike (w or B): 25 m
- Fill friction angle (ϕ): 32°
- Adherence ratio (r_s): 0.5
- Overlying load (P_0): 0
- Factor of Safety (FS): 1

1. Determine the Critical Sliding Plane Angle (α)

First, we find the critical sliding plane angle using the internal friction angle of the backfill ¹ :

$$\alpha = 45^\circ + \frac{\phi}{2} = 45^\circ + \frac{32^\circ}{2} = 61^\circ$$

2. Check the Aspect Ratio Condition

We must check the aspect ratio to determine if this is a High Aspect Ratio (HAR) or Low Aspect Ratio (LAR) slope ².

$$\frac{H}{B} = \frac{50}{25} = 2.0$$

$$\tan(\alpha) = \tan(61^\circ) \approx 1.804$$

Since $H/B \geq \tan \alpha$ ($2.0 \geq 1.804$), the slope qualifies as a **High Aspect Ratio (HAR)** slope, meaning the sliding plane will intersect the back wall rather than the top of the backfill ².

3. Calculate the Modified Height (H^*)

For a HAR slope, the modified height H^* is calculated as ³:

$$H^* = H - \frac{w \tan \alpha}{2} = 27.45 \text{ m}$$

4. Calculate the Required Cohesion (c)

Using the Li & Aubertin equation for a HAR slope 3 :

$$c = \frac{(P_0 + \gamma H^*)/2}{\left[\left(FS - \frac{\tan \phi}{\tan \alpha} \right) \sin(2\alpha) \right]^{-1} + \frac{r_s H^*}{L}} = 116.85 \text{ kPa}$$

Let's break down the calculation into the numerator and the two terms of the denominator:

- **Numerator:** $(0 + 21.2 \times 27.45)/2 = 581.94/2 = \mathbf{290.97 \text{ kPa}}$
- **Denominator Term 1:** $\left[\left(1 - \frac{\tan(32^\circ)}{\tan(61^\circ)} \right) \sin(122^\circ) \right]^{-1}$
 - $\tan(32^\circ) \approx 0.6249$
 - $\tan(61^\circ) \approx 1.8040$
 - $1 - (0.6249/1.8040) = 1 - 0.3464 = 0.6536$
 - $\sin(122^\circ) \approx 0.8480$
 - $[0.6536 \times 0.8480]^{-1} = [0.5543]^{-1} \approx \mathbf{1.804}$

5. Calculate the Required Unconfined Compressive Strength (UCS)

Assuming the failure of the backfill follows the Coulomb criterion, the expression for required UCS is 4 :

$$UCS = 2c \tan \left(45^\circ + \frac{\phi}{2} \right)$$

$$UCS = 2(116.85) \times \tan(61^\circ)$$

$$UCS = 233.7 \times 1.804 \approx \mathbf{421.6 \text{ kPa}}$$

4. Unit weight of fill of 21.2 kN/m³, slope height 30 m, thickness of slope as 20 m, slope width along strike 25, fill friction angle as 32 degree, adherence ratio as 0.5, no overlying load, using Li & Aubertin solution determine the required strength of backfill, consider FS 1.

- Unit weight of fill (γ): 21.2 kN/m³
- Slope height (H): 30 m
- Thickness of stope (L , perpendicular to strike): 20 m
- Slope width along strike (B or w): 25 m
- Fill friction angle (ϕ): 32°
- Adherence ratio (r_s): 0.5
- Overlying load (P_0): 0
- Factor of Safety (FS): 1

1. Determine the Critical Sliding Plane Angle (α)

First, calculate the critical sliding plane angle using the internal friction angle of the backfill 6 :

$$\alpha = 45^\circ + \frac{\phi}{2} = 45^\circ + \frac{32^\circ}{2} = 61^\circ$$

2. Check the Aspect Ratio Condition

We must check the aspect ratio (H/B) to determine if this is a High Aspect Ratio (HAR) or Low Aspect Ratio (LAR) stope 2 3 .

$$\frac{H}{B} = \frac{30}{25} = 1.2$$

$$\tan(\alpha) = \tan(61^\circ) \approx 1.804$$

Since $H/B < \tan \alpha$ ($1.2 < 1.804$), the stope qualifies as a **Low Aspect Ratio (LAR)** stope 2 . This means the potential sliding plane will intersect the top of the backfill rather than the back wall 4 .

3. Calculate the Required Cohesion (c)

For a LAR slope, the required cohesion is calculated using Li & Aubertin's specific equation for this condition ⁵ :

$$c = \frac{P_0 + \frac{\gamma H}{2}}{2 \left[\left(FS - \frac{\tan \phi}{\tan \alpha} \right) \sin(2\alpha) \right]^{-1} + \frac{r_s H}{L}} = 72.97 \text{ kPa}$$

Let's break down the calculation into the numerator and the two terms of the denominator:

- **Numerator:** $0 + \frac{21.2 \times 30}{2} = \frac{636}{2} = 318 \text{ kPa}$
- **Denominator Term 1:** $2 \times \left[\left(1 - \frac{\tan(32^\circ)}{\tan(61^\circ)} \right) \sin(122^\circ) \right]^{-1}$
 - $\tan(32^\circ) \approx 0.6249$
 - $\tan(61^\circ) \approx 1.8040$
 - $1 - (0.6249/1.8040) = 1 - 0.3464 = 0.6536$
 - $\sin(122^\circ) \approx 0.8480$

4. Calculate the Required Unconfined Compressive Strength (UCS)

Assuming the failure of the backfill follows the Coulomb criterion, the expression for the required UCS is ⁴ :

$$UCS = 2c \tan \left(45^\circ + \frac{\phi}{2} \right)$$

$$UCS = 2(72.97) \times \tan(61^\circ)$$

$$UCS = 145.94 \times 1.804 \approx 263.27 \text{ kPa}$$